

Predictive Model Plan

Customer Delinquency Prediction

1. Model Logic

Using ChatGPT, I generated a predictive model using logistic regression to estimate the likelihood of a customer becoming delinquent. The model uses key features such as Credit_Utilization, Missed_Payments, Income, Debt_to_Income_Ratio, and Account_Tenure to predict a binary outcome: 1 if the customer is likely to become delinquent, and 0 otherwise.

Pseudo-code:

1. Load dataset
2. Select features: ['Credit_Utilization', 'Missed_Payments', 'Income', 'Debt_to_Income_Ratio', 'Account_Tenure']
3. Define target variable: 'Delinquent_Account'
4. Split data into training and testing sets
5. Fit logistic regression model
6. Predict and evaluate using classification metrics

The model follows the steps outlined below:

1. Input customer data including income, credit score, credit utilization, missed payments, debt-to-income ratio, loan balance, age, employment status, and credit card type.
2. Handle missing values using median imputation for numerical variables and mode imputation for categorical variables.
3. Encode categorical variables using one-hot encoding.
4. Standardize numerical features to ensure consistent scale across variables.
5. Train a logistic regression model with L2 regularization to predict the probability of delinquency.
6. Generate a probability score between 0 and 1 for each customer indicating delinquency risk.

The model is designed to support early risk identification and enable proactive financial decision-making.

2. Justification for Selected Model

Logistic Regression: was selected as the primary model due to its strong alignment with financial risk modeling requirements. I chose logistic regression because it is widely used for binary classification problems and is highly interpretable. In financial services, we consider:

Accuracy: The model provides consistent and reliable performance on structured financial data and generalizes well to unseen data when properly tuned.

Transparency: Logistic regression offers clear interpretability through model coefficients, allowing stakeholders to understand how each variable influences delinquency risk.

Ease of Implementation: The model is computationally efficient, easy to deploy, and straightforward to monitor and retrain over time.

Relevance for Financial Prediction: Logistic regression is widely accepted in credit risk and banking environments due to its stability and auditability.

Business Suitability: For Geldium's use case, the model balances predictive power with regulatory compliance, customer fairness, and operational simplicity.

3. Evaluation Strategy

To evaluate the model, I would use accuracy, precision, recall, F1 score, ROC and AUC. The model's performance will be evaluated using these metrics to ensure balanced and ethical decision-making.

Evaluation Metrics:

- **Accuracy** Measures overall correctness of predictions.
- **Precision** Assesses how many predicted delinquent customers are truly delinquent.
- **Recall** Evaluates the model's ability to correctly identify actual delinquent customers.
- **F1 Score** Balances precision and recall for imbalanced data.
- **ROC-AUC** Measures the model's ability to distinguish between delinquent and non-delinquent customers.

Interpretation Strategy:

Recall is prioritized to minimize missed delinquent customers, while precision is monitored to avoid excessive false positives.

Bias Detection and Reduction:

Model performance will be evaluated across income and employment groups to identify disparities. Threshold tuning and probability calibration will be applied to reduce bias.

Ethical Considerations:

The model avoids direct use of sensitive attributes and focuses on financial behavior. Predictions are used as decision-support tools rather than sole decision-makers, ensuring fairness and responsible use.